



CHEMISTRY HSSC-I

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent.

Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات ای مطبوعہ پر کرنا غیر مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لگنے کی اجازت نہیں ہے۔ لید پینسل کا استعمال ممنوع ہے۔

Version No.				
3	0	0	5	1

ROLL NUMBER						

0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
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1	1	1	1	1	1	1
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9	9	9	9	9	9	9

Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پر کریں۔

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The type of intermolecular forces present between Hydrochloric acid molecules is:	Ionic bond	Metallic Force	Dipole Dipole Forces	Hydrogen Bonding	☐	☐	☐	☐
2. The reactant that determines the maximum yield of a chemical reaction is called the:	Excess reactant	By-product	Catalyst	Limiting reactant	☐	☐	☐	☐
3. If 2 moles of H_2 react with 32g of O_2 , the number of moles of H_2O formed will be: $2H_2 + O_2 \rightarrow 2H_2O$	1	3	2	4	☐	☐	☐	☐
4. Which property of water is explained by hydrogen bonding?	Low boiling point	High surface tension	High vapor pressure	Low viscosity	☐	☐	☐	☐
5. The enthalpy change when 1 mole of a substance is completely burned in oxygen is called:	Heat of formation	Heat of combustion	Heat of atomization	Heat of neutralization	☐	☐	☐	☐
6. Which substance has zero value for their standard enthalpy of formation?	H_2O	CO_2	Fe_2O_3	O_3	☐	☐	☐	☐
7. A catalyst increases the rate of reaction by:	Increasing activation energy	Decreasing temperature	Changing equilibrium constant	Lowering activation energy	☐	☐	☐	☐
8. If an element "X" has 1 st ionization energy $495 kJ/mol$ and 2 nd ionization energy $4560 kJ/mol$, this indicates that there is/are valence electrons.	1	2	3	5	☐	☐	☐	☐
9. A conjugate acid–base pair differs by:	One neutron	One electron	One proton	One molecule	☐	☐	☐	☐
10. The number of valence electrons in an atom can be deduced from its:	Atomic mass	Density	Group number	Isotopic composition	☐	☐	☐	☐
11. Which of the following has no specific functional group?	Ethene	Propane	Butanol	Acetic Acid	☐	☐	☐	☐
12. The reaction of methane with chlorine in the presence of sunlight is an example of:	Electrophilic addition	Free radical substitution	Nucleophilic substitution	Elimination	☐	☐	☐	☐
13. According to Markovnikov's rule, the stability of carbocations formed during electrophilic addition is explained by:	Resonance effect	Hydrogen bonding	Inductive effect of alkyl groups	Dispersion forces	☐	☐	☐	☐
14. Which reagent is used to distinguish aldehydes from ketones?	Bayer's reagent	Lindler catalyst	Tollens' reagent	Haber reagent	☐	☐	☐	☐
15. Aldehydes can be oxidized to:	Carboxylic acids	Ketones	Alcohols	Esters	☐	☐	☐	☐
16. Which sub-shell can hold a maximum of 14 electrons?	f	d	p	s	☐	☐	☐	☐
17. Which of the following halogens exists in solid state at room temperature?	I_2	Cl_2	Br_2	F_2	☐	☐	☐	☐

—1HA-II 25005—

SUPPLEMENTARY TABLE

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40

**CHEMISTRY HSSC-I****Time allowed: 2:35 Hours****Total Marks Sections B and C: 68****SECTION – B (Marks 42)****Q. 2 Answers the following parts briefly.****(14 x 3 = 42)**

(i)	Write the electronic configuration of phosphorus ($^{31}_{15}P$) and Carbon ($^{12}_6C$) according to Hund's rule.	1.5 + 1.5	OR	Write any three differences between sigma (σ) and pi (π) bonds.	1x3
(ii)	How azimuthal quantum number helps to determine the number of electrons in d- sub shell? Explain briefly.	03	OR	What mass of Mg is needed to produce 100cm^3 of H_2 at STP? $Mg + 2HCl \rightarrow MgCl_2 + H_2$	03
(iii)	Why Bond energy in F-F is lower than in Cl-Cl? Although the size of F_2 is smaller than Cl_2 .	03	OR	Why Heat of vaporization ΔH_v is always greater than ΔH_f Heat of fusion?	03
(iv)	Determine the number of protons, neutrons and electrons in Na^+ and Cl^- ions.	1.5 + 1.5	OR	Briefly explain any three uses of sulphur in industry or in daily life.	03
(v)	Consider the following substances to identify the type of Bond (ionic, polar or non-polar covalent bond) on the basis of Electronegativity difference $O_2(O: 3.44), KCl(K: 0.82), HCl(Cl: 3.16)(H: 2.20)$	03	OR	Write K_{sp} expression for the following molecules: a. CaF_2 b. $MgSO_4$	1.5 + 1.5
(vi)	Determine the limiting reactant when 10g of Al reacts with 20g of Fluorine according to given equation: $2Al + 3F_2 \rightarrow 2AlF_3$	03	OR	Justify the followings: a. Atomic size decreases along the period. b. Metallic character increases along the Group.	1.5 + 1.5
(vii)	Differentiate between crystalline and amorphous solids with examples.	03	OR	Write the General formulas of Primary, Secondary and Tertiary amines.	1x3
(viii)	Differentiate between Exothermic and Endothermic reactions with suitable example in each case.	1.5 + 1.5	OR	What is Catenation? How is it important in organic Chemistry?	1+2
(ix)	Draw Energy cycle according to Hess's law for the given reaction: $C + O_2 \rightarrow CO_2 \quad \Delta H^\circ = -393.5\text{kJ/mol}$ If this reaction occurs in two steps: $C + \frac{1}{2}O_2 \rightarrow CO \quad \Delta H^\circ_1 = -110.52\text{kJ/mol}$ $CO + \frac{1}{2}O_2 \rightarrow CO_2 \quad \Delta H^\circ_2 = -282.98\text{kJ/mol}$	03	OR	Calculate K_p if the following concentrations are found at Equilibrium in given reaction: $2SO_2 + O_2 \rightleftharpoons 2SO_3$ $[SO_2] = 0.59M \quad [O_2] = 0.05M \quad \text{Temperature} = 450^\circ C$ $[SO_3] = 0.259M$	03
(x)	$N_2 + 3H_2 \rightleftharpoons 2NH_3$ This reaction is at equilibrium. What will happen if: a. Ammonia is removed from the mixture b. More nitrogen is added in the mixture.	1.5 + 1.5	OR	Write chemical equation for the preparation of alkyl halides: a. From Alkenes b. From Alkanes	1.5 + 1.5
(xi)	Derive the relation of Acid dissociation constant K_a for the following dissociation equation for an acid (HX) $HX + H_2O \rightleftharpoons H_3O^+ + X^-$	03	OR	How Lucas test can differentiate between Primary, Secondary and Tertiary alcohols?	1x3
(xii)	Enlist the factors which affect the reactivity of alkyl halides. Write correct order of reactivity for them.	2+1	OR	Describe the use of 2,4-dinitrophenylhydrazine to detect presence of carbonyl compounds.	03
(xiii)	Write any three differences between boiling and evaporation.	1x3	OR	Complete the given equations by identifying the major products: a. $CH_3 - Cl + KCN \rightarrow$ b. $CH_3 - Cl + NH_3 \rightarrow$	1.5 + 1.5
(xiv)	Describe the mechanism of free radical substitution in alkanes using example of methane with chlorine.	03	OR	How can Iodoform test be used to distinguish between: a. Methanol and Ethanol b. Formaldehyde and Acetaldehyde	1.5 + 1.5

SECTION – C (Marks 26)**Attempt the following questions.**

Q.3	What is hybridization? Explain hybridization with molecular diagrams in Methane (CH_4) and Boron Tri Fluoride (BF_3).	1+3 +3	OR	State Markovnikov's rule. Write mechanism of reaction and identify the major product when propene reacts with Hydrogen Bromide.	2+4 +1
Q.4	Calculate the mass of Calcium Nitride prepared from 54.9 g of Ca and 43.2 g of N_2 . $3Ca + N_2 \rightarrow Ca_3N_2$ If actual yield is 0.13 moles of Ca_3N_2 then calculate the percentage Yield.	4+3	OR	Complete the following chemical equations by identifying the major products obtained for the preparation of alkyl halides: a. $CH_3CH_2OH \xrightarrow[HCl]{ZnCl_2}$ b. $CH_3CH_2OH \xrightarrow[Pyridine]{SOCl_2}$ c. $CH_3CH_2OH \xrightarrow{PBr_3}$	3+2 +2
Q.5	Use the Maxwell-Boltzmann distribution curve to explain the effect of temperature on the rate of a reaction with the help of General graphical representation.	3+3	OR	How are following conversions done? a. Ethanol into Acetaldehyde b. Ethanol into Ethyl Acetate Ester c. Ethanol into Ethene	2+2 +2
Q.6	What is meant by Nucleophilic addition reactions? Write mechanism of the reaction when HCN reacts with Formaldehyde and NaOH act as a base.	2+4	OR	State Le-Chatelier's Principal. How it helps to obtain maximum product in: a. Exothermic reactions b. Endothermic reactions	2+2 +2

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